

Graph product of multifuncoids

OPG-37389

L1 - TRACTABLE

PARTIALLY SOLVED

Topology

ORIGINAL PROBLEM STATEMENT

Conjecture Let F is a family of multifuncoids such that each F_i is of the form $\lambda j \in N(i) : \mathfrak{F}(U_j)$ where $N(i)$ is an index set for every i and U_j is a set for every j . Let every $F_i = E^* f_i$ for some multifuncoid f_i of the form $\lambda j \in N(i) : \mathfrak{P}(U_j)$ regarding the filtrator $(\prod_{j \in N(i)} \mathfrak{F}(U_j); \prod_{j \in N(i)} \mathfrak{P}(U_j))$. Let H is a graph-composition of F (regarding some partition G and external set Z). Then there exist a multifuncoid h of the form $\lambda j \in Z : \mathfrak{P}(U_j)$ such that $H = E^* h$ regarding the filtrator $(\prod_{j \in Z} \mathfrak{F}(U_j); \prod_{j \in Z} \mathfrak{P}(U_j))$.

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